

Automating Daily Backups: How to Schedule Oracle APEX Application Exports



Vikas Pandey • March 18, 2020 • APEX

How to Automate Daily Exports for Your Oracle APEX Application



Today, I'm excited to share a solution for automating the export of Oracle APEX applications, eliminating the need for manual exports. This tutorial will guide you through a streamlined process to set up automatic daily exports of your APEX applications. Follow these simple steps to ensure your application data is backed up consistently and efficiently.

1. Create a Directory and Set Permissions

- **Create Directory:** First, you need to create a directory for storing the exported files. You can name it `APEX_TEMP` or any name of your choice.
- **Grant Permissions:** Ensure that the Oracle database user running the export process has read and write permissions on this directory.

```
CREATE DIRECTORY apex_temp AS 'C:\APEX_TEMP';
GRANT READ, WRITE ON DIRECTORY apex_temp TO admin;
```

[COPY CODE](#)

Replace `C:\APEX_TEMP` with the actual path where you want to store the exported files, and `APEX_TEMP` with the appropriate Oracle APEX user.

2. Create a Table for Storing Application Export Details

Objective: Set up a table to hold the export data for your Oracle APEX applications. This table will store the exported application content in a CLOB (Character Large Object) format along with a reference to the application ID.

```
CREATE TABLE export_clob
(
    app_export    CLOB,           -- Column to store the exported application data
    app_id        VARCHAR2(5)    -- Column to store the application ID
);
```

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`app_export`: This column will store the actual exported application data as a CLOB.

`app_id`: This column will store the ID of the application being exported, which helps in identifying and managing multiple applications.

- **Objective:** Develop a PL/SQL procedure to automate the export of your Oracle APEX application and save it into the directory you created. This procedure will handle the export process and store the export data in the `export_clob` table.

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```

CREATE OR REPLACE PROCEDURE blob_to_file (p_dir      IN VARCHAR2,
                                           p_file     IN VARCHAR2,
                                           p_blob      IN BLOB)

AS
    l_file      UTL_FILE.file_type;
    l_buffer     RAW (32767);
    l_amount     BINARY_INTEGER := 32767;
    l_pos        INTEGER := 1;
    l_blob_len   INTEGER;
BEGIN
    l_blob_len := DBMS_LOB.getlength (p_blob);
    l_file :=
        UTL_FILE.fopen (p_dir,
                        p_file,
                        'WB',
                        32767);

    WHILE l_pos < l_blob_len
    LOOP
        DBMS_LOB.read (p_blob,
                        l_amount,
                        l_pos,
                        l_buffer);
        UTL_FILE.put_raw (l_file, l_buffer, TRUE);
        l_pos := l_pos + l_amount;
    END LOOP;

    UTL_FILE.fclose (l_file);
EXCEPTION
    WHEN OTHERS
    THEN
        IF UTL_FILE.is_open (l_file)
        THEN
            UTL_FILE.fclose (l_file);
        END IF;

        RAISE;
END blob_to_file;

```

Check out blob to clob and clob to blob conversions

- [Clob to Blob using apex webservice](#)
- [Blob to Clob using DBMS_LOB](#)
- [CLOB to BLOB using PLSQL](#)
- [BLOB to CLOB using APEX_WEB_SERVICE](#)
- Now Need to write the SQL file into the Directory "APEX_TEMP " using the previous Procedure.



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```
CREATE OR REPLACE PROCEDURE write_to_dir
IS
    CURSOR get_clob
    IS
        SELECT app_export, app_id, clob_to_blob (app_export) b_file
        FROM export_clob;
BEGIN
    FOR i IN get_clob
    LOOP
        blob_to_file (
            'APEX_TEMP ',
            TO_CHAR (SYSDATE, 'DDMONYYHH24MISS') || '_' || i.app_id || '.sql',
            i.b_file);
    END LOOP;
END write_to_dir;
/
```

- Automate the process of exporting an Oracle APEX application, saving the export data to a database table, and writing the file to a specified directory.



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```

CREATE OR REPLACE PROCEDURE export_application
IS
    l_files    apex_t_export_files;
    contents   CLOB;

    CURSOR get_app
    IS
        SELECT application_id
            FROM apex_applications
           WHERE application_id = 101;
BEGIN
    DELETE FROM export_clob;

    FOR i IN get_app
    LOOP
        l_files :=
            apex_export.get_application (p_application_id => i.application_id);
        contents := l_files (1).contents;

        INSERT INTO export_clob (app_id, app_export)
            VALUES (i.application_id, contents);
    END LOOP;

    COMMIT;

    BEGIN
        write_to_dir;
        COMMIT;
    END;
END;

```

- OR



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```

CREATE OR REPLACE PROCEDURE export_application
IS
    CURSOR get_app
    IS
        SELECT application_id
          FROM apex_applications
         WHERE application_id IN (101, 108);
BEGIN
    DELETE FROM export_clob;

    FOR i IN get_app
    LOOP
        INSERT INTO export_clob (app_id, app_export)
            VALUES (
                i.application_id,
                wwv_flow_utilities.export_application_to_clob (
                    i.application_id));
    END LOOP;

    COMMIT;

    BEGIN
        write_to_dir;
        COMMIT;
    END;
END;

```

- To set up a job scheduler in the Oracle Database that automatically exports an APEX application daily, follow these steps. This process involves creating a job that executes the `EXPORT_APPLICATION` procedure at a specified time each day.

```

CREATE SEQUENCE exp_apax_seq START WITH 1 INCREMENT BY 1;

BEGIN
    DBMS_SCHEDULER.create_job (
        job_name          => 'VIKAS' || exp_apax_seq.NEXTVAL,
        job_type           => 'PLSQL_BLOCK',
        job_action          => 'begin
                                EXPORT_APPLICATION;
                                end;',
        start_date          => SYSDATE, --+1/24/59, -- sysdate + 1 minute
        repeat_interval     => 'freq=MINUTELY;interval=5',
        enabled             => TRUE,
        auto_drop           => TRUE,
        comments            => 'Application Versoin control in Oracle Apex');
END;

```

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Vikas Pandey

I am a skilled developer with over 8+ years of experience in Oracle APEX, Oracle Database, and APEX Themes and Plugins. As an Oracle ACE Associate, I focus on delivering innovative solutions that enhance user experiences and streamline business processes. My expertise includes database design and performance optimization, allowing me to tackle complex challenges effectively. I am passionate about leveraging technology to create impactful solutions and actively seek opportunities for growth and collaboration.

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